

Bachelor in Computer Applications

Course Title: **Operating System**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. What are the different types of operating system? Differentiate between Real time and Batch OS.
2. What is kernel? Explain the process of system call with suitable diagram.
3. What is Resource Allocation Graph (RAG)? What are the necessary conditions for deadlock? Explain. Determine if the system is safe from deadlock or not using Banker's Algorithm for given condition. [Note: Consider process sequence as: P1, P2, P3, P4, P5] Need Matrix Allocation Matrix MAX Resource Matrix
Process ABC Process ABC ABC P1143 P131117810 P2321 P2131 P3020 P3411 P4324 P4010 P5523 P5201
4. What is an OS? Explain 3-state model of process with neat and clean diagram.
5. What is race condition? Explain critical problem solution using lock variable.
6. What is memory compaction? Explain best fit memory allocation with suitable example.
7. What is RAID? Explain Level-2 and Level-3 RAID.
8. How threads differ from processes? Explain user level thread and kernel level thread.
9. Consider the following page reference string: 3,2,1,3,4,2,3,9,7,4,2,9,8,3. Calculate the total number of page faults for Optimal and LRU page replacement algorithms using 4 page frame
10. Explain the working mechanism of DMA.
11. What is TLB? Explain the importance of TLB in conversion of logical address to physical address.
12. Consider 500 GB hard drive with 5 KB block size. Calculate the size of file allocation table if entry for each block needs 4 bytes.